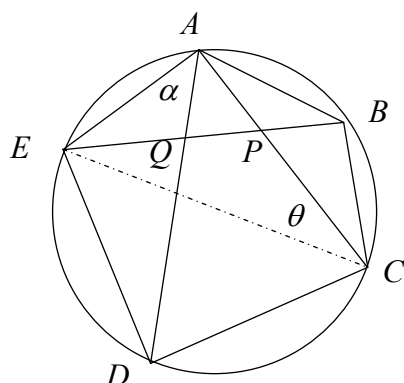


Question 1.	Marks
(a) Sketch $y = 2 - e^{-x}$, showing all essential detail.	2
(b) Evaluate $\lim_{x \rightarrow 0} \left(\frac{3x}{\sin 5x} \right)$.	2
(c) Find the acute angle formed between the two lines: $y = 2$ and $y = x\sqrt{3} - 2$.	2
(d) Solve $2 \cos \theta + \sqrt{3} = 0$ for $0 \leq \theta \leq 2\pi$.	2
(e) Express $2 \sin x + \cos x$ in the form $R \sin(x + \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.	2
(f) Find the number of ways in which the letters of the word EPSILON can be	
(i) arranged in a circle.	1
(ii) arranged in a row so that the three vowels are all next to each other.	2
(g) (i) Show that $x - 2$ is a factor of the polynomial expression: $x^3 - 3x^2 + 4$.	1
(ii) Hence express $x^3 - 3x^2 + 4$ as a product of three linear factors.	1
Question 2. [START A NEW PAGE]	
(a) Show that $\tan 2x + \tan x \equiv \frac{\sin 3x}{\cos 2x \cos x}$.	2
(b) Find $\frac{dy}{dx}$ in the following:	
(i) $y = \sec(x^3)$.	1
(ii) $y = 10x - x \ln x$.	2
(c) Given that $\cos 75^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}$ and using the double angle results, or otherwise, find the exact value of $\cos \left(37\frac{1}{2}^\circ \right)$.	2

Question 2 continued over the page

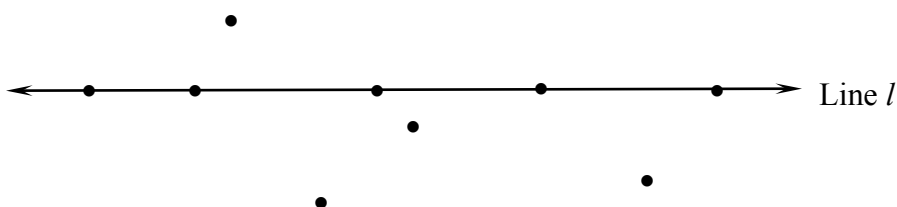
- (c) $ABCDE$ is a pentagon inscribed in a circle, where $AB = AE$, BE meets AC and AD at P and Q respectively.



Not to scale

Given $\angle ACE = \theta$ and $\angle DAE = \alpha$,

- (i) Copy the diagram onto your writing paper and show that $\angle AEB = \theta$, give reasons. 2
- (ii) Hence, or otherwise show that $CPQD$ is a cyclic quadrilateral. 3
- (d) The diagram shows 9 points lying in the plane, 5 of which lie on the line l . The remaining 4 points do not lie on line l and no other set of 3 points is collinear.



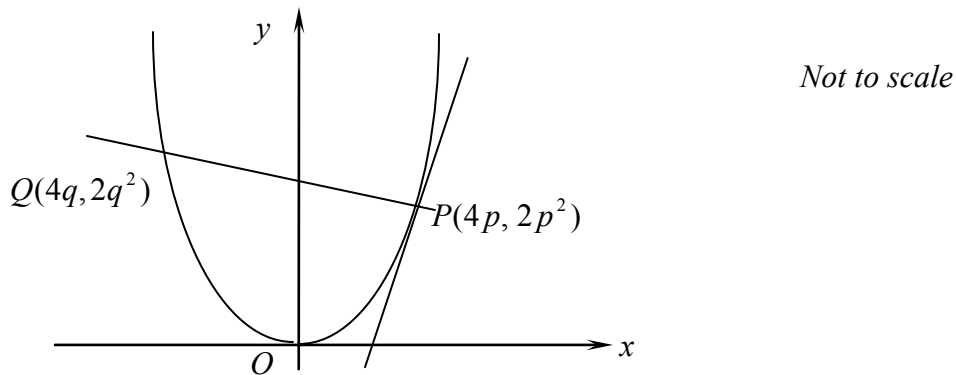
- (i) How many sets of 3 points can be chosen from the 5 points lying on l ? 1
- (ii) How many distinct triangles can be formed using any three of the 9 points as vertices? 2

Question 3.

[START A NEW PAGE]

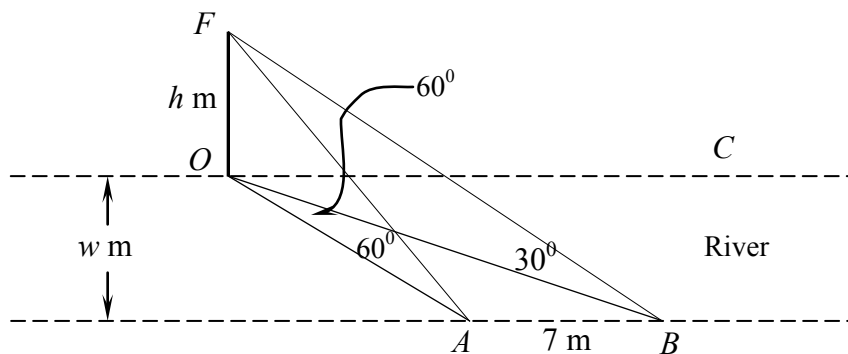
Marks

- (a) Point $P(4p, 2p^2)$ lies on the parabola shown. The normal at P intersects the parabola at the point $Q(4q, 2q^2)$.



- (i) Show that the Cartesian equation for this parabola is $x^2 = 8y$. 1
- (ii) Show that the equation of the normal at P is given by:

$$x + py = 2p^3 + 4p.$$
 3
- (iii) Hence, or otherwise, show that $p + q = -\frac{2}{p}$, where $p \neq 0$. 2
- (iv) Find the coordinates of the midpoint M of chord PQ . 1
- (v) Hence find the locus of M as points P and Q move on the parabola. 3
- (b) A river has level parallel riverbanks OC and AB of width w metres. OF is a vertical flagpole of height h metres which stands with its base O on the edge of riverbank OC . Positions A and B are two points on the other riverbank such that $AB = 7$ metres and $\angle AOB = 60^\circ$. The angle of elevation to the top of the flagpole from A and B are 60° and 30° respectively, as shown below.



- (i) Show that the height of the flagpole is $\sqrt{21}$ metres. 3
- (ii) By finding the area of $\triangle AOB$, or otherwise, find the width of the river. 2

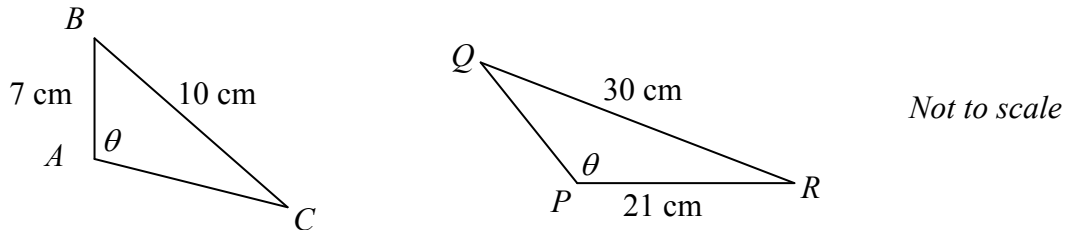
Question 4.

[START A NEW PAGE]

Marks

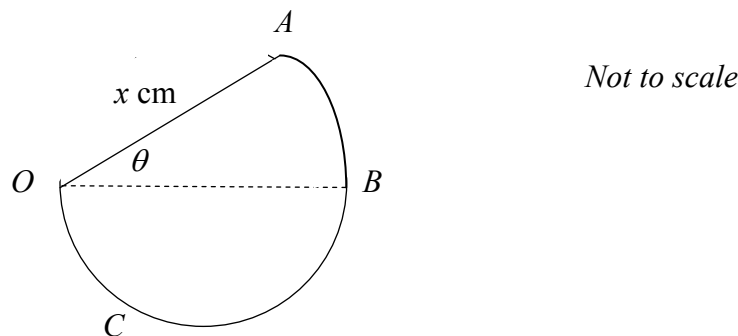
- (a) For what value(s) of k is the line $y = 12x + k$ a tangent to $y = x^3$. **3**

- (b) Given the triangles ABC and PQR where $\angle BAC = \angle RPQ = \theta$ and $\theta \geq 90^\circ$.



Copy the diagrams onto your writing paper and using the Sine rule, prove that $\triangle ABC \parallel \triangle PRQ$. **3**

- (c) A cam for a motor is made with a cross-section as shown below.



The cam's cross-section consists of a semi-circle OCB and the sector of a circle OAB of centre O with a radius of x cm and with an included angle $BOA = \theta$.

- (i) Determine the formula for the perimeter P cm of the cam $OABC$, in terms of x and θ . **2**

- (ii) Given that the area of the cross-section of the cam is 1 cm^2 , show that **2**

$$x^2 = \frac{8}{\pi + 4\theta}.$$

- (iii) Hence, or otherwise, show that the perimeter P is given by: **1**

$$P = \frac{(\pi + 2 + 2\theta)\sqrt{2}}{\sqrt{\pi + 4\theta}}.$$

- (iv) Determine the value of θ for least perimeter of the cam, (answer correct to the nearest minute). **4**

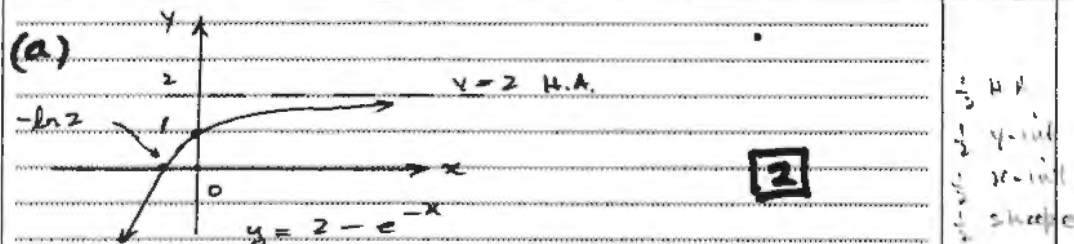
YEAR 11 MATH EXT 1, T3, 2007
PRELIMINARY SOLUTIONS

MATHEMATICS Extension 1: Question 1

Suggested Solutions

Marks

Marker's Comments



(b) $\lim_{x \rightarrow 0} \frac{3x}{\sin 5x} = \lim_{x \rightarrow 0} \frac{3x(5x)}{5x \sin(5x)}$

$= \frac{3}{5} \lim_{x \rightarrow 0} \frac{(5x)}{\sin(5x)}$

$= \frac{3}{5} \times 1 = \frac{3}{5}$

2

(c) METHOD 1 $y_1 = 2$ $m_1 = 0$

$y_2 = x\sqrt{3} - 2$ $m_2 = \sqrt{3}$

$\tan \theta = \left| \frac{0 - \sqrt{3}}{1 + 0 \times \sqrt{3}} \right| = \frac{\sqrt{3}}{1}$

$\therefore \angle \theta = 60^\circ$

2

(d) $2 \cos \theta = -\sqrt{3}$

$\cos \theta = -\frac{\sqrt{3}}{2}$

$\angle \theta = \pi - \frac{\pi}{6}$ or $\pi + \frac{\pi}{6}$

$\therefore \angle \theta = \frac{5\pi}{6}$ or $\frac{7\pi}{6}$

2

(e) $R \sin(x + \alpha) = 2 \sin x + \cos x$

$R \sin x \cos \alpha + R \cos x \sin \alpha = 2 \sin x + \cos x$

$\therefore R \cos \alpha = 2 \dots (1)$ $\cos \alpha = \frac{2}{R}$

$R \sin \alpha = 1 \dots (2)$ $\sin \alpha = \frac{1}{R}$

$\therefore R = \sqrt{2^2 + 1^2} = \sqrt{5}$ $\therefore \alpha = \cos^{-1} \frac{2}{\sqrt{5}} / \sin^{-1} \frac{1}{\sqrt{5}} / \tan^{-1} \frac{1}{2} = 0.4636 \dots$

$\therefore \sqrt{5} \sin(x + \alpha)$, where $\alpha = \dots$

2

(f) E I O P S L N $n = 7$

(i) No of ways = $\frac{7!}{7} = 6! = 720$

(ii) $\boxed{E|I|O|P|S|L|N}$ No of ways = $(4 \times 3!) \times 4! = 576$

2

(g) (i) $P(x) = x^3 - 3x^2 + 4$

$P(2) = 8 - 3 \times 4 + 4 = 0$ $\therefore$ by Factor thm $(x-2)$ is a factor $\checkmark$

2

(ii) $x^3 - 3x^2 + 4 = (x-2)(x^2 - x - 2)$

$= (x-2)(x-2)(x+1)$

2

using LONG Division

$x-2 \overline{) x^3 - 3x^2 + 0x + 4}$

$\underline{x^3 - 2x^2}$

$0 - x^2 + 0x$

$\underline{-x^2 + 2x}$

$0 -2x + 4$

$\underline{-2x + 4}$

0

YEAR 11 MATH EXT 1, T3 2007
PRELIMINARY SOLUTIONS

MATHEMATICS Extension 1 : Question 2

Suggested Solutions

Marks

Marker's Comments

(a) LHS $\tan 2x + \tan x = \frac{\sin 2x}{\cos 2x} + \frac{\sin x}{\cos x}$ $\frac{1}{2} + \frac{1}{2}$

$$= \frac{\sin 2x \cos x + \cos 2x \sin x}{\cos 2x \cos x}$$

[2]

$$= \frac{\sin(2x+x)}{\cos 2x \cos x} = \frac{\sin 3x}{\cos 2x \cos x}$$

(b) Using $\cos 2\theta = 2\cos^2\theta - 1$ or
 $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$
 $\cos^2 37\frac{1}{2}^\circ = \frac{1}{2}(1 + \frac{\sqrt{6}-\sqrt{2}}{4})$ **[2]**
 $= \frac{1}{2} \cdot \frac{4 + \sqrt{6} - \sqrt{2}}{4}$
 $\cos 37\frac{1}{2}^\circ = \pm \sqrt{\frac{4 + \sqrt{6} - \sqrt{2}}{8}}$
 as $\cos 37\frac{1}{2}^\circ > 0$ $\cos 37\frac{1}{2}^\circ = \frac{\sqrt{4 + \sqrt{6} - \sqrt{2}}}{2\sqrt{2}}$

(c) (i) $\frac{dy}{dx} = \sec(x^3) \tan(x^3) \cdot 3x^2$ **[1]**
 (ii) $\frac{dy}{dx} = 10 - (1 \cdot \ln x + x \cdot \frac{1}{x})$ **[2]**
 $= 9 - \ln x$

(d)

(i) 1. $\angle ABE = \theta$ (Angles in the same segment are equal when standing on same arc AE)
 2. $\angle AEB = \theta$ (Equal angles opposite equal sides AE, AB) **[2]**

(ii) 1. $\angle ECD = \alpha$ (Angles in same segment are equal when standing on same arc DE)
 2. $\angle AQB = \theta + \alpha$ (Exterior angle of $\triangle AQE$ equals sum of the two opposite interior angles)
 3. As $\angle PCQ = \angle AQB = \theta + \alpha$
 $\therefore PCQR$ is a cyclic quad. as the exterior angle at Q equals the interior opposite angle C .

(e)

(i) No. of ways = ${}^5C_2 = 10$ **[1]**

(ii) No. of Δ s = ${}^9C_3 - {}^5C_3 = {}^9C_3 - 10$ **[2]**
 $= 84 - 10$
 $= 74$

Ans! For 9/3
 Ans! For - (i)

MATHEMATICS Extension 1 : Question 3

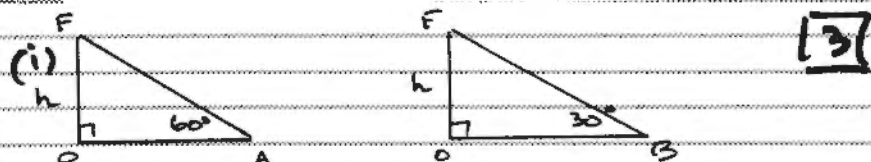
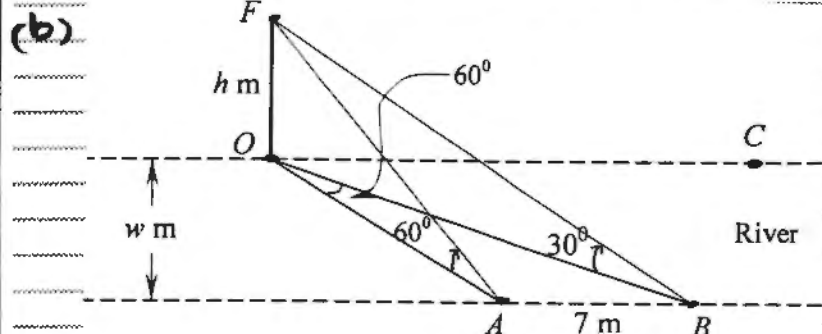
Suggested Solutions	Marks	Marker's Comments
(a) (i) $x = 4p \Rightarrow p = \frac{x}{4} \therefore y = 2 \times \frac{x^2}{16}$ $y = \frac{x^2}{8}$ $\therefore 8y = x^2$ 1		$(4p)^2 = 8 \times (2p^2)$ <u>Aw!</u> For showing
(ii) $y = \frac{x^2}{8}$ $y' = \frac{x}{4}$ 2 Gradient of Tangent at P $m_T = \frac{4p}{4} = p$ " " Normal at P $m_N = -\frac{1}{p}$ $\therefore$ Equation of Normal at P: $y - 2p^2 = -\frac{1}{p}(x - 4p)$ $py - 2p^3 = -x + 4p$ $\therefore x + py = 2p^3 + 4p$ <u>qed.</u>	✓ ✓ ✓	<u>Aw!</u> For getting to next st <u>Aw!</u> For $m_N = -\frac{1}{p}$ <u>Aw!</u> For getting to result.
(iii) $m_{PQ} = -\frac{1}{p}, p \neq 0$ $\frac{2p^2 - 2q^2}{4p - 4q} = -\frac{1}{p}$ 2 $\frac{2(p-q)(p+q)}{4(p-q)} = -\frac{1}{p}$ $\frac{p+q}{2} = -\frac{1}{p}, q \neq p$ $\therefore p+q = -\frac{2}{p}; p \neq 0$ p & q satisfies (ii) $4q + p \cdot 2q^2 = 2p^3 + 4p$ $2q + pq^2 = p^3 + 2p$ $pq^2 - p^3 = 2p - 2q$ $p(q^2 - p^2) = 2(p - q)$ $p(q - p)(q + p) = 2(p - q)$ $\therefore q + p = -\frac{2}{p}, q \neq p \neq 0$		<u>Aw!</u> For using p & q in m_{PQ} or (ii) <u>Aw!</u> For getting to result via factorization
(iv) $M = \left(\frac{4p+4q}{2}, \frac{2p^2+2q^2}{2} \right) = (2(p+q), p^2+q^2)$ 1		
Let $M = (x, y)$ be the general point on the required locus (v) $\therefore x \equiv 2(p+q)$ ---- (1) $y \equiv p^2 + q^2$ ---- (2) 3 $p+q = -\frac{2}{p}$ ---- (3) $p+q = \frac{x}{2} = -\frac{2}{p} \therefore p = -\frac{4}{x}$ Now $p^2 + q^2 \equiv (p+q)^2 - 2pq$ $y = \left(\frac{x}{2}\right)^2 - 2pq$ but (3) $p^2 + pq = -2$ $pq = -2 - p^2$ $= -2 - \left(-\frac{4}{x}\right)^2$ $= -2 - \frac{16}{x^2}$ $\therefore y = \frac{x^2}{4} - 2\left(-2 - \frac{16}{x^2}\right)$ $y = \frac{x^2}{4} + \frac{32}{x^2} + 4, x \neq 0$ this is the locus of M	1 1 1	OR $y = p^2 + q^2$ but $q = -\frac{2}{p} - p$ and $p = -\frac{4}{x}$ $q = \frac{x}{2} + \frac{4}{x}$ $\therefore y = \left(-\frac{4}{x}\right)^2 + \left(\frac{x}{2} + \frac{4}{x}\right)^2$ $= \frac{16}{x^2} + \frac{x^2}{4} + \frac{22}{x^2} + 8$ <u>Aw!</u> For getting $p = -\frac{4}{x}$ <u>Aw!</u> For using $p^2 + q^2$... <u>Aw!</u> For getting locus

MATHEMATICS Extension 1 : Question 3

Suggested Solutions

Marks

Marker's Comments



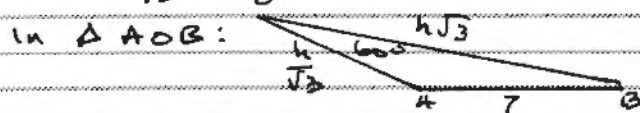
$$\tan 60^\circ = \frac{h}{OA}$$

$$\tan 30^\circ = \frac{h}{OB}$$

$$OA = \frac{h}{\tan 60^\circ} = h \tan 30^\circ \quad OB = \frac{h}{\tan 30^\circ} = h \tan 60^\circ$$

$$OA = \frac{h}{\sqrt{3}}$$

$$OB = h\sqrt{3}$$



$$7^2 = 3h^2 + \frac{h^2}{3} - 2 \times \frac{h}{\sqrt{3}} \times \frac{h\sqrt{3}}{\sqrt{3}} \cos 60^\circ$$

$$7^2 = h^2 \left[3 + \frac{1}{3} - 2 \times 1 \times \frac{1}{2} \right]$$

$$= \frac{7}{3} h^2$$

$$h^2 = \frac{3 \times 7^2}{7} = 21 \Rightarrow h = \sqrt{21}$$

$\therefore$ height is $\sqrt{21}$ m

(ii) Area $\triangle AOB = \frac{1}{2} \times \frac{h}{\sqrt{3}} \times h\sqrt{3} \times \sin 60^\circ = \frac{1}{2} \times 7 \times w$

$$\text{i.e. } \frac{1}{2} \times h^2 \times \frac{\sqrt{3}}{2} = \frac{1}{2} \times 7 \times w$$

$$21 \times \frac{\sqrt{3}}{2} = 7w$$

$$\therefore w = 3\sqrt{3}$$

$$\therefore \text{width is } 3\sqrt{3} \text{ m.}$$

OTHERWISE:

Aw! For getting

$$OA = \frac{h}{\sqrt{3}} \text{ and } OB = h\sqrt{3}$$

Aw! For using cosine rule correctly in $\triangle AOB$

Aw! For getting to $\sqrt{21}$ correctly

Aw! For using areas correctly

Aw! For getting $w = 3\sqrt{3}$

MATHEMATICS Extension 1 : Question 4

Suggested Solutions

Marks

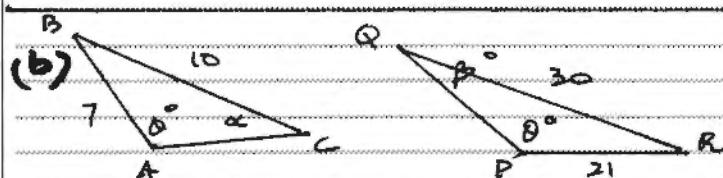
Marker's Comments

(a) $y = x^3$, $y = 12x + k$
 $\frac{dy}{dx} = 3x^2$
 $\therefore k = 12$
 $\therefore 3x^2 = 12$
 $x^2 = 4$
 $x = \pm 2$
 $\therefore y = (\pm 2)^3 = \pm 8$
 $\Rightarrow \pm 8 = 12(\pm 2) + k = \pm 24 + k$
 $\therefore k = \mp 16$



[3]

01: $x^3 - 12x - k = 0$
 α, α and β
 $2\alpha + \beta = 0 \dots (1)$
 $\alpha^2 + 2\alpha\beta = -12 \dots (2)$
 $\alpha^2\beta = k \dots (3)$
 $(1) \Rightarrow 4\alpha^2 + 2\alpha\beta = 0$
 $(2) \quad \frac{\alpha^2 + 2\alpha\beta = -12}{3\alpha^2 = 12}$
 $\alpha = \pm 2 \rightarrow \beta = \mp 4$
 $k = (\pm 2)^2(\mp 4) = \mp 16$



[3]

In ΔABC and PRQ
 $\frac{\sin \alpha}{7} = \frac{\sin \theta}{10} = \frac{\sin B}{AC}$
and $\frac{\sin \beta}{21} = \frac{\sin \theta}{30} = \frac{\sin R}{PQ}$
 $\therefore \frac{\sin \alpha}{7} = \frac{\sin \theta}{10} \Rightarrow \frac{21 \sin \theta}{30} = \sin \beta$
 $\therefore \sin \alpha = \frac{7}{18} \sin \theta = \frac{21 \sin \theta}{30} = \sin \beta$
 $\therefore \alpha = \beta$ or $\alpha = 180 - \beta$
but $\alpha > 90$ $\therefore \alpha = \beta$ only

1. $\frac{AB}{PR} = \frac{7}{21} = \frac{1}{3}$ and $\frac{BC}{QR} = \frac{10}{30} = \frac{1}{3}$

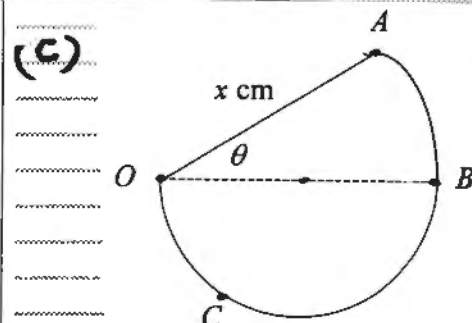
2. $\angle C = \angle Q = \alpha$

$\therefore \Delta ABC \parallel \Delta PRQ$ (sides about equal angles are in same ratio)

NB: once shown $\alpha = \beta$ only, then
1. $\angle A = \angle P = \theta$
2. $\angle C = \angle Q = \alpha$
 $\therefore \Delta ABC \parallel \Delta PRQ$
(corresponding angles)

Ans 2 For showing $\alpha = \beta$ only

Ans 1 For proving Δ s similar.



$r = \frac{x}{2}$ for circle

(i) $P = OA + \text{arc } AB + BCO(\text{arc})$
 $= x + x\theta + \frac{1}{2} \times 2\pi \times \frac{x}{2}$
 $P = x + x\theta + \frac{\pi x}{2}$

[2]

(ii) Area = $\frac{1}{2} x^2 \theta + \frac{1}{2} \times \pi \times (\frac{x}{2})^2 = 1$

i.e. $\frac{1}{2} x^2 \theta + \frac{\pi x^2}{8} = 1$

$x^2(4\theta + \pi) = 8$

$\therefore x^2 = \frac{8}{\pi + 4\theta}$

[2]

MATHEMATICS Extension 1 : Question 4

Suggested Solutions

Marks

Marker's Comments

$$(iii) \quad x = \frac{\sqrt{8}}{\sqrt{\pi+4\theta}} = \frac{2\sqrt{2}}{\sqrt{\pi+4\theta}} \quad \text{from (ii)}$$

$\frac{1}{2}$

$$\therefore P = x + x\theta + \frac{\pi}{2}x$$

$$= \frac{x}{2} [2 + 2\theta + \pi]$$

$\square$

$$= \frac{1}{2} \times \frac{2\sqrt{2}}{\sqrt{\pi+4\theta}} [\pi + 2 + 2\theta]$$

$\frac{1}{2}$

$$\therefore P = \frac{[\pi + 2 + 2\theta]\sqrt{2}}{\sqrt{\pi+4\theta}} \quad \text{qed.}$$

$$(iv) \quad \frac{dP}{d\theta} = \sqrt{2} \left\{ \frac{2\sqrt{\pi+4\theta} - (\pi+2+2\theta)\frac{1}{2}(\pi+4\theta)^{-1/2} \times 4}{(\pi+4\theta)} \right\}$$

$$= \sqrt{2} \left\{ \frac{2\sqrt{\pi+4\theta} - 2(\pi+2+2\theta)}{\sqrt{\pi+4\theta}} \right\}$$

$\square$

$$= \sqrt{2} \left\{ \frac{2(\pi+4\theta) - 2(\pi+2+2\theta)}{(\pi+4\theta)^{3/2}} \right\}$$

$$= \sqrt{2} \left\{ \frac{2\pi + 8\theta - 2\pi - 4 - 4\theta}{(\pi+4\theta)^{3/2}} \right\}$$

$$\frac{dP}{d\theta} = \frac{[4\theta - 4]\sqrt{2}}{(\pi+4\theta)^{3/2}} = \frac{4(\theta-1)\sqrt{2}}{(\pi+4\theta)^{3/2}}$$

2

For possible max/min values of P to occur $\frac{dP}{d\theta} = 0$

$$\therefore \theta = 1 \quad P = \sqrt{2} \times \sqrt{\pi+4} = \dots$$

$\frac{1}{2}$

Ans 2 For using Quotient rule correctly

Ans $\frac{1}{2}$ For θ

TEST

θ	$\frac{1}{2}$	1	$1\frac{1}{2}$
$\frac{dP}{d\theta}$	$\frac{4(-1)\sqrt{2}}{(\pi+2)^{3/2}}$ -1.25	0	$\frac{4(1)\sqrt{2}}{(\pi+6)^{3/2}}$ 0.94
	$\backslash$	-	$\swarrow$

1

Ans 1 For testing θ properly

$\therefore$ a relative min T.P at $\theta = 1$

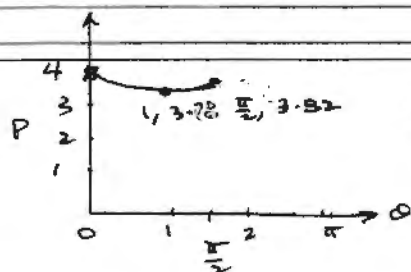
and since P continuous for $0 \leq \theta \leq \frac{\pi}{2}$

$\therefore$ Least perimeter when $\theta = 1$

i.e. $\angle Q = 57^\circ 18'$

$\frac{1}{2}$

Ans 1 For answers



basic sketch.

$$\theta = 0 \quad P = \frac{(\pi+2)\sqrt{2}}{\sqrt{\pi}} \div 4.10$$

$$\theta = 1 \quad P = \sqrt{2\pi+8} \div 3.78$$

$$\theta = \frac{\pi}{2} \quad P = \frac{(2+2\pi)\sqrt{2}}{\sqrt{3\pi}} \div 3.82$$